CHM 306: Transition Metal Chemistry (Mid Sem 1)

Date: 04/02/2026, Time: 5.00-6.00 PM

(Maximum Marks: 20)

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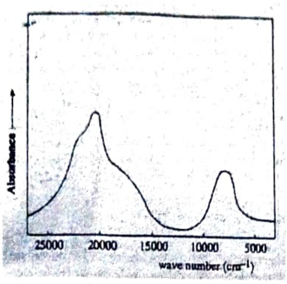
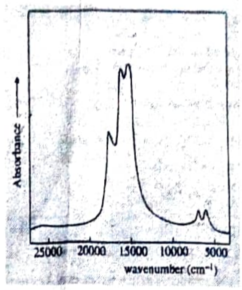
1. The gross features of the electronic spectrum of $[Co(H_2O)_6]^{2+}$ (left) and $CoCl_4^{2-}$ (right) are shown below. **3+3+1**

(a) In $[Co(H_2O)_6]^{2+}$ assign the corresponding electronic transitions for the bands at 8000, 16000, and 19400 cm^{-1} using Orgel diagram.

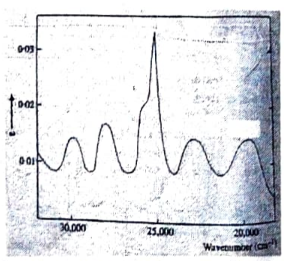
(b) In $CoCl_4^{2-}$ the lowest energy band lies in the infrared region (3300 cm^{-1}, not shown in the figure). Assign the corresponding electronic transitions to the other two bands at 5800 and 15000 cm^{-1} using Orgel diagram.

(c) Which complex will show intense color in solution and why?

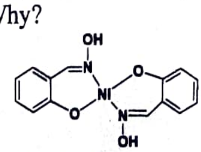




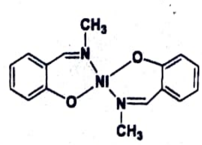
2. The gross features of the electronic spectrum of $[Mn(H_2O)_6]^{2+}$ is shown below. Assign the corresponding electronic transitions for the bands at 18900, 23100, and 24970, 25300, 28000 and 29700 cm^{-1} using Orgel diagram. **4**



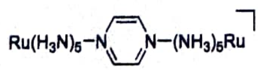
3. The below mentioned nickel(II)salicylaldoximate complex is a colorless crystalline solid but in pyridine solution it becomes green. Why? **1**



4. The below mentioned nickel(II) complex is diamagnetic in solid but in $CHCl_3$ solution, it shows paramagnetic behavior. Why? **1**



5. Mention the type of charge-transfer interaction in the below-mentioned compounds for their intense colors. **4**



$[Pt(NH_3)_4][PtCl_4]$

MoO_3 Sb_2S_3 $Cr(CO)_6$ MnO_4^- $Fe_4[Fe(CN)_6]_3$ $Cs_2Au_2Br_6$

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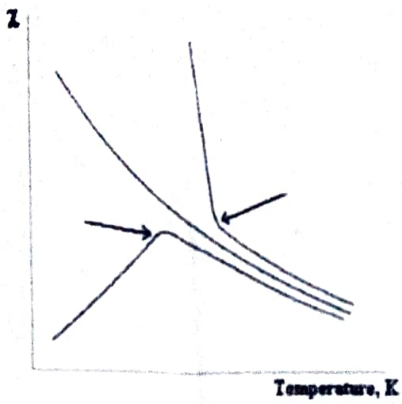
6. The below-mentioned plot shows the variation in magnetic susceptibility with temperature. Mention the types of magnetic materials that show such behavior and also mention the names of the deviation points shown in blue arrow. 3



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No marks will be given for selecting the correct option without proper justification.

1. [2]

$$[Cr(H_2O)_6]^{2+} + bpy \xrightleftharpoons{K_1} [Cr(bpy)(H_2O)_4]^{2+} + 2\,H_2O$$
$$[Cr(bpy)(H_2O)_4]^{2+} + bpy \xrightleftharpoons{K_2} [Cr(bpy)_2(H_2O)_2]^{2+} + 2\,H_2O$$
$$[Cr(bpy)_2(H_2O)_4]^{2+} + bpy \xrightleftharpoons{K_3} [Cr(bpy)_4]^{2+} + 2\,H_2O$$

1. The correct order of the formation constants for the above-mentioned substitution reaction is

(a) $K_1>K_2>K_3$ (b) $K_1<K_2>K_3$ (c) $K_3>K_1>K_2$ (d) $K_1<K_2<K_3$ (e) $K_1>K_3<K_2$

2. [3]

$$[M(H_2O)_6]^{2+} + en \xrightleftharpoons{K_1} [M(en)(H_2O)_4]^{2+} + 2\,H_2O$$
$$[M(en)(H_2O)_4]^{2+} + en \xrightleftharpoons{K_2} [M(en)_2(H_2O)_2]^{2+} + 2\,H_2O$$
$$[M(en)_2(H_2O)_2]^{2+} + en \xrightleftharpoons{K_3} [M(en)_3]^{2+} + 2\,H_2O$$

$$M = Cu^{2+}, Co^{2+}, Ni^{2+}$$

2. The correct order of the stepwise formation constants and overall formation constant (β) for the above-mentioned substitution reactions will be

(a) $K_{1(Cu)}>K_{1(Ni)}>K_{2(Co)}$
$K_{2(Cu)}>K_{2(Ni)}>K_{2(Co)}$
$K_{3(Cu)}>K_{3(Ni)}>K_{3(Co)}$
$\beta_{(Cu)}>\beta_{(Ni)}>\beta_{(Co)}$

(b) $K_{1(Co)}>K_{1(Ni)}>K_{2(Cu)}$
$K_{2(Co)}>K_{2(Ni)}>K_{2(Cu)}$
$K_{3(Co)}>K_{3(Ni)}>K_{3(Cu)}$
$\beta_{(Co)}>\beta_{(Ni)}>\beta_{(Cu)}$

(c) $K_{1(Cu)}>K_{1(Ni)}>K_{2(Co)}$
$K_{2(Cu)}>K_{2(Ni)}>K_{2(Co)}$
$K_{3(Ni)}<K_{3(Co)}<K_{3(Cu)}$
$\beta_{(Cu)}>\beta_{(Co)}>\beta_{(Ni)}$

(d) $K_{1(Ni)}>K_{1(Co)}>K_{2(Cu)}$
$K_{2(Ni)}>K_{2(Co)}>K_{2(Cu)}$
$K_{3(Cu)}>K_{3(Ni)}>K_{3(Co)}$
$\beta_{(Ni)}>\beta_{(Co)}>\beta_{(Cu)}$

(e) $K_{1(Ni)}>K_{1(Co)}>K_{2(Cu)}$
$K_{2(Ni)}>K_{2(Co)}>K_{2(Cu)}$
$K_{3(Co)}>K_{3(Ni)}>K_{3(Cu)}$
$\beta_{(Ni)}>\beta_{(Co)}>\beta_{(Cu)}$

(f) $K_{1(Cu)}>K_{1(Ni)}>K_{2(Co)}$
$K_{2(Cu)}>K_{2(Ni)}>K_{2(Co)}$
$K_{3(Ni)}>K_{3(Co)}>K_{3(Cu)}$
$\beta_{(Cu)}>\beta_{(Ni)}>\beta_{(Co)}$

(g) $K_{1(Cu)}<K_{1(Ni)}<K_{2(Co)}$
$K_{2(Cu)}<K_{2(Ni)}>K_{2(Co)}$
$K_{3(Ni)}>K_{3(Co)}>K_{3(Cu)}$
$\beta_{(Cu)}<\beta_{(Ni)}>\beta_{(Co)}$

(h) $K_{1(Co)}>K_{1(Ni)}>K_{2(Cu)}$
$K_{2(Co)}>K_{2(Ni)}>K_{2(Cu)}$
$K_{3(Ni)}>K_{3(Co)}>K_{3(Cu)}$
$\beta_{(Co)}>\beta_{(Ni)}>\beta_{(Cu)}$

3. [2]

$$[Co(CN)_5(H_2O)]^{2-} + X^- \xrightleftharpoons{\beta} [Co(CN)_5(X)]^{3-} + H_2O$$
$$[Co(NH_3)_5(H_2O)]^{3+} + X^- \xrightleftharpoons{\beta} [Co(NH_3)_5(X)]^{2+} + H_2O$$

3. The correct order of halide ligands with respect to the increasing order of formation constant (log β) for the above-mentioned reactions will be:

(a) $Cl^->Br^->I^-$; $I^->Br^->Cl^-$
(b) $Cl^->Br^-<I^-$; $I^->Br^-<Cl^-$
(c) $Cl^->Br^-<I^-$; $I^->Br^-<Cl^-$
(d) $I^->Br^->Cl^-$; $I^-<Br^-<Cl^-$
(e) $Br^->I^->Cl^-$; $Br^->I^->Cl^-$
(f) $I^->Br^->Cl^-$; $I^->Br^->Cl^-$

4. [2]

$$cis\text{-}[Co(en)_2(OH)Cl]^+ + H_2O \xrightarrow{K_1} cis\text{-}[Co(en)_2(OH)H_2O]^{2+} + Cl^-$$
$$cis\text{-}[Co(en)_2(NH_3)Cl]^{2+} + H_2O \xrightarrow{K_2} cis\text{-}[Co(en)_2(NH_3)H_2O]^{3+} + Cl^-$$

4. The correct order of the rate constants for the above-mentioned substitution reactions will be

(a) $K_1>K_2$ (b) $K_1<K_2$ (c) $K_1 \approx K_2$ (d) cannot be predicted

5. [2+1]

$$cis\text{-}[Co(en)_2(OH)Cl]^+ + H_2O \xrightarrow{K_1} cis\text{-}[Co(en)_2(OH)H_2O]^{2+} + Cl^-$$
$$trans\text{-}[Co(en)_2(OH)Cl]^+ + H_2O \xrightarrow{K_2} cis + trans\text{-}[Co(en)_2(OH)H_2O]^{2+} + Cl^-$$

5a. The correct order of the rate constants for the above-mentioned substitution reactions will be

(a) $K_1>K_2$ (b) $K_1<K_2$ (c) $K_1 \approx K_2$ (d) cannot be predicted

5b. %Yield of the isomers between cis and trans in the second reaction will be

(a) cis<trans (b) cis>trans (c) 2:1 trans:cis (d) cis≈trans

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cis-$[Co(en)_2(NH_3)Cl]^{2+} + H_2O \xrightarrow{K_1}$ *cis*-$[Co(en)_2(NH_3)H_2O]^{3+} + Cl^-$ *trans*-$[Co(en)_2(NH_3)Cl]^{+} + H_2O \xrightarrow{K_2}$ *trans*-$[Co(en)_2(NH_3)H_2O]^{2+} + Cl^-$ 6. The correct order of the rate constants for the above-mentioned substitution reactions will be (a) $K_1>K_2$ (b) $K_1<K_2$ (c) $K_1\approx K_2$ (d) cannot be predicted	2
$[Fe(bpy)_3]^{2+} + 2\,H_2O \xrightarrow[H^+]{K_1} [Fe(bpy)_2(H_2O)_2]^{2+} + bpyH^+$ $[Fe(bpy)_3]^{2+} + 2\,H_2O \xrightarrow{K_2} [Fe(bpy)_2(H_2O)_2]^{2+} + bpy$ $[Fe(phen)_3]^{2+} + 2\,H_2O \xrightarrow[H^+]{K_3} [Fe(phen)_2(H_2O)_2]^{2+} + phenH^+$ 7. The correct order of the rate constants for the above-mentioned reactions will be (a) $K_1>K_2>K_3$ (b) $K_1\approx K_3>K_2$ (c) $K_1>K_3>K_2$ (d) $K_2\approx K_1<K_3$ (e) $K_2>K_3<K_1$ (e) $K_3>K_1>K_2$	3
$[Co(en)_2(NO_2)Cl]^{+} + OH^- \xrightarrow{K_1} [Co(en)_2(NO_2)OH]^{+} + Cl^-$ $[Co(en)_2Cl_2]^{+} + OH^- \xrightarrow{K_2} [Co(en)_2(Cl)OH]^{+} + Cl^-$ 8. The correct order of the rate constants for the above-mentioned reactions will be (a) $K_1>K_2$ (b) $K_1<K_2$ (c) $K_1\approx K_2$ (d) cannot be predicted	3
en: Ethylenediamine, bpy: 2,2'-Bipyridine, phen: 1,10 Phenanthroline	

CHM 306: Transition Metal Chemistry (End Sem)

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(Maximum Marks: 50)

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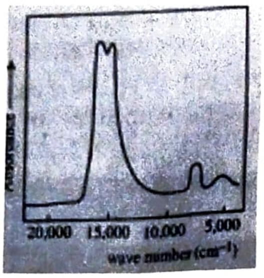
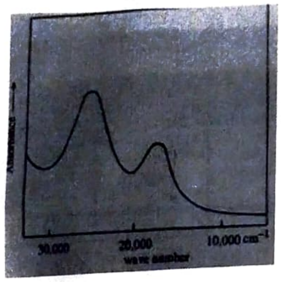
1. The gross features of the electronic spectrum of $[NiCl_4]^{2-}$ (left) and $[V(H_2O)_6]^{3+}$ (right) are shown below. **3+3+2**

(a) In $[NiCl_4]^{2-}$ assign the corresponding electronic transitions for the bands at 4000, 7500, and 15000 cm^{-1} using Orgel diagram.

(b) In $[V(H_2O)_6]^{3+}$ assign the corresponding electronic transitions for the bands at 17200 and 25000 cm^{-1} using Orgel diagram.

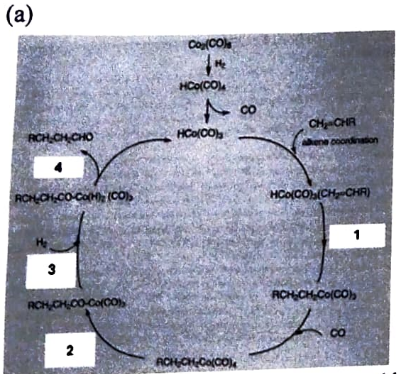




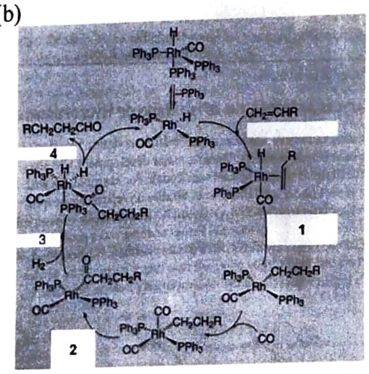
(c) With the help of Orgel diagram, show the possible electronic transitions in CrF_6^{3-} complex.

2. Mention the type of reactions involved in the below mentioned catalytic pathways. Also mention the name of the overall process for all the transformations. **3.5+1**

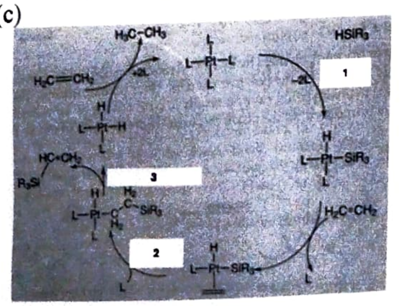
(a)



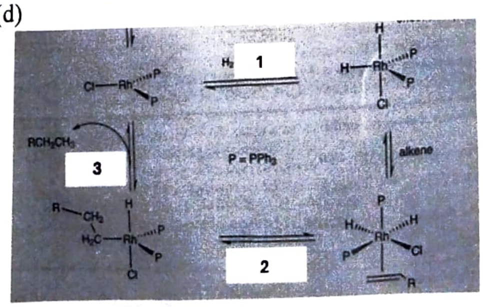
(b)



(c)



(d)



3. Mention the final product of the reaction along with the mechanistic pathways involved in the reaction. **2.5**

$$CO + CH_3OH \xrightarrow[\substack{\text{HI} \\ \text{(catalytic amount)} \\ 180\ °C}]{[Rh(CO)_2I_2]^-}$$

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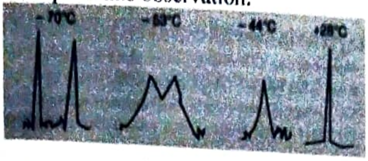
4. The ^{1}H-NMR spetrum of $Cp_2Fe_2(CO)_4$ (where Cp= cyclopentadienyl) at different temperature (-70 to +28 °C) was shown below. Why there are two peaks at low temperature but only one peak was observed at higher temperature. Explain this observation. **2**



5. Mention the correct order of increasing carbonyl frequency of $Ni(CO)_4$, $Ni(CO)_3(P(OPh)_3)$, $Ni(CO)_3(PMe_3)$ and $Ni(CO)_3(PF_3)$ and why? **1**

6. (a) Which metal carbonyl(s) can be isolobal to CH_3^- and justify. $CpCr(CO)_2$, $Fe(CO)_4$, $Co(CO)_3$, $Cr(CO)_5$, $Ni(CO)_3$, $Fe(CO)_5$, $Cr(CO)_4$ **1+1**
(b) Which Cp-M (Cp= cyclopentadienyl, M = Metal) can be isolobal to a BH fragment, and justify? CpMn, CpCo, CpGe, CpRu, CpFe.

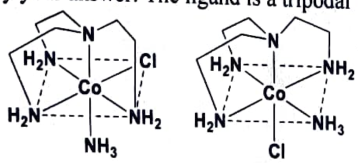
7. Using the 18-electron rule as a guide, determine the value of 'x' in the following complexes: **2**

(a) $[Ni(CO)_3(NO)]^x$ (consider NO as linear) (b) $[\eta^6\text{-}(C_6H_6)Mn(CO)_2(CH_3)]^x$
(c) $[\eta^5\text{-}Cp(CO)_2Fe(PhC \equiv CH)]^x$ (d) $[(Cp)Fe(CO)(PPh_3)(CF_2)]^x$

8. For the following compounds, calculate: (a) the total number of M–M bonds in each molecule, (b) the number of M–M bonds each metal makes with the other metal, and (c) draw the most appropriate structure in each case. **10**

(a) $(\mu\text{-}H)_2\{\eta^5\text{-}Cp(NO)WH\}_2$ (NO is linear)
(b) $(\mu\text{-}Br)_2\{Mn(CO)_4\}_2$
(c) $\mu\text{-}CO\text{-}\mu\text{-}CH_2\{\eta^5\text{-}CpRh\}_2$
(d) $(\mu\text{-}CO)\text{-}[\eta^4\text{-}(C_4H_4)Fe(CO)]_2$
(e) $[Cp(CO)Ir(\mu\text{-}CO)_2Re(CO)Cp]$

9. Comment on the rate and product stereochemistry for the base hydrolysis (OH^-) of the below mentioned two isomers. Justify your answer. The ligand is a tripodal 'tren' ligand. **3**



10. The correct order (with justification) of the rates for the below-mentioned aquation reactions will be **2**

$$[Co^{III}(NH_3)_5Cl]^{2+} + H_2O \xrightarrow{k_1} [Co^{III}(NH_3)_5H_2O]^{3+} + Cl^-$$
$$[Co^{III}(NH_3)_4Cl_2]^{+} + H_2O \xrightarrow{k_2} [Co^{III}(NH_3)_4(Cl)H_2O]^{2+} + Cl^-$$

(a) $k_1 \approx k_2$ (b) $k_1 >> k_2$ (c) $k_2 > k_1$ (d) $k_1 > k_2$ (e) cannot be determined.

11. The correct order (with justification) of the rates for the below-mentioned reactions will be: (pn: 1, 2 diaminopropane; en: ethylenediamine) **2**

$$[Co^{III}(pn)_2Cl_2]^{+} + H_2O \xrightarrow{k_1} [Co^{III}(pn)_2(Cl)H_2O]^{2+} + Cl^-$$
$$[Co^{III}(en)_2Cl_2]^{+} + H_2O \xrightarrow{k_2} [Co^{III}(en)_2(Cl)H_2O]^{2+} + Cl^-$$

(a) $k_1 \approx k_2$ (b) $k_1 > k_2$ (c) $k_2 > k_1$ (d) cannot be determined.

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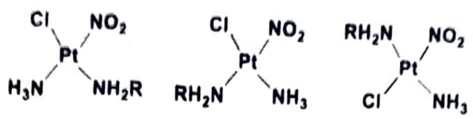
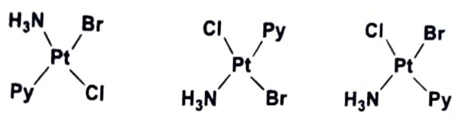
12. Write the synthetic route for the below-mentioned three isomers from $PtCl_4^{2-}$ 3



13. Write the synthetic route for the below-mentioned three isomers from $PtCl_4^{2-}$ 3



14. Justify the observation in the product formations of the following electron transfer reactions. 2

$$[Co^{III}(NH_3)_5(SCN)]^{2+} + [Cr(H_2O)_6]^{2+} \xrightarrow{H_3O^+} [Co(H_2O)_6]^{2+} + [Cr(NCS)(H_2O)]^{2+} + [Cr(SCN)(H_2O)]^{2+}$$

$$[Co^{III}(NH_3)_5(NCS)]^{2+} + [Cr(H_2O)_6]^{2+} \xrightarrow{H_3O^+} [Co(H_2O)_6]^{2+} + [Cr(NCS)(H_2O)]^{2+}$$

15. Mention the correct order of the rates and the final products for the below-mentioned electron transfer reactions. Justify your answer. 3